The residue-field case: reducing the proof to what the local theorem uses

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Status. The diagonal, general flatness transfer and field regularity in Sections 2–5 are now checked in Unit, including the finite span, an actual finite basis and a perfect trace pairing. The finite-field primitive-element theorem, the small-coefficient-field branch, selected relative bases and selected simple-root equations also check. The large-field diagonal search and both field endpoints now check: finite presentation and formal unramifiedness give a finite product of separable monic quotients, without a prime parameter, as well as a separable monogenic neighbourhood of a specified decidable prime. The product proof simplifies the large-field argument by using a divided difference and trace duality directly. The remaining local work is the connection supplying all local-base inputs. The starting point is Lombardi's *Un théorème décisif* [1] and Lombardi–Quitté [2]. The suggested reorganisation uses the flatness and idempotent-kernel arguments already checked in `sag/etale-flat-unramified`.

Its transfer theorem for a standard *étale* source has now been generalised to a finitely presented, formally unramified source in `sag/etale-field-transfer`. The diagonal construction is in `sag/etale-field-diagonal`; the field application is in `sag/etale-field-regularity`. The combined theorem is `UnramifiedFieldStructure` in `sag/etale-field`; no finite basis or intermediate theorem is assumed in its input. The selected-prime conclusion is `UnramifiedFieldSelected` in `sag/etale-field-complete`; it constructs the basis, trace pairing, diagonal and primitive factor from those same hypotheses and the specified prime. The full product conclusion is `UnramifiedFieldProduct` in `sag/etale-field-product`; it constructs both directions of the algebra isomorphism and uses no specified prime or factorisation oracle.

1 Start from the two actual outputs

Let k be a nonzero discrete field and C a finitely presented k-algebra, unramified in the sense of uniqueness of lifting across square-zero ideals.

First, the ZMT application needs that C is finite over k. A finite basis is a useful stronger output, provided it is constructed, not inferred from finite generation alone.

Second, the local structure proof has a *specified decidable prime* $\mathfrak{p}$ of C. It needs only a monogenic separable factor which contains that prime. A sufficient field-level output is

$$e^2 = e, \qquad e \notin \mathfrak{p}, \qquad C_e = k[x], \qquad P(x) = 0, \qquad P'(x) \in C_e^{\times},$$

with $P \in k[T]$ monic. An isomorphism with $k[T]/(P)$ is more than the downstream proof requires; a separable annihilating polynomial and generation suffice. Nor does it need a decomposition into fields, irreducibility tests, or all factors at once.

There is a further finite-interface simplification. The current `Introduction.Generates` asks for polynomial representations of every $b \in B$, with untruncated witnesses. The only actual consumption, after the change $x \mapsto x - 1$, is at the finitely many module generators `bs i`: see `Introduction.seg`, the declarations `Ci` and `e-gen`. The field bridge should therefore supply merely the representations of these finitely many elements. Finite choice then suffices. This refactoring now checks in `sag/etale-field-local-interface`: private variants of the introduction and local proof consume just these finitely many representations. `SelectedFlatLocal` takes one factor outside the specified prime, without requiring a global product decomposition. The earlier files are unchanged.

2 A common certificate: the diagonal idempotent

Lemma 1. *Let $R \to S$ be a finitely presented, formally unramified algebra. Put $D = S \otimes_R S$ and $J = \ker(\mu : D \to S)$. There exists $\epsilon \in D$ such that*

$$\mu(\epsilon) = 1, \qquad (s \otimes 1)\epsilon = (1 \otimes s)\epsilon \quad (s \in S).$$

Moreover $\epsilon^2 = \epsilon$.

Proof. If $x_1, \ldots, x_n$ generate S, the differences $x_i \otimes 1 - 1 \otimes x_i$ generate J. Consider the two maps $S \rightrightarrows D/J^2$. They become equal in $D/J \simeq S$; the kernel J/J^2 of this last quotient is square-zero. Uniqueness of lifting makes the two maps equal already in D/J^2. Thus each difference belongs to J^2 and $J = J^2$.

Apply the determinant lemma for a finitely generated idempotent ideal. It supplies ϵ with $\epsilon J = 0$ and $1 - \epsilon \in J$. These are precisely the two required identities, and $\epsilon(1 - \epsilon) = 0$ gives idempotence. □

This proof starts from the existing `FormallyEtale.Unique` interface. It needs no construction of the full module of Kähler differentials or an equivalence with the Jacobian criterion. The quotient and determinant infrastructure already exists. The underlying diagonal argument is also recorded in [4].

3 Extend the checked flatness argument

Proposition 2. *Suppose $R \to S$ has the certificate of Lemma 1. For any R-algebra map $S \to T$, if T is flat over R, it is flat over S.*

Proof. The algebra $D_T = S \otimes_R T$ is flat over S by flat base change. Map the second factor of ϵ to T, obtaining ϵ_T. The multiplication map $\mu_T : D_T \to T$ has an S-linear section

$$t \longmapsto \epsilon_T(1 \otimes t).$$

Linearity follows from the equalisation identity, and the composite with μ_T is the identity since $\mu_T(\epsilon_T) = 1$. A module retract of a flat module is flat. □

The proof is the same averaging mechanism used in the checked standard étale transfer theorem. There, synthetic division constructs the averaging element. Here, formal unramifiedness and the determinant lemma construct it.

Both extensions now check. `DiagonalCertificate` constructs the diagonal certificate from `Unique`. `FlatPresentedCocone` iterates polynomial base change and simultaneous principal quotients, then transports the source along its presentation isomorphism. It supplies a flat algebra with the two structure maps and the mapping-out laws needed for averaging; it does not require a finite presentation of T. `FlatDiagonalTransfer` combines these constructions.

4 Finiteness without the derivation argument

Throughout this note, *zero-dimensional* means the algebraic property

$$\forall x\ \exists N \in \mathbb{N},\ b \qquad x^N(1 - xb) = 0.$$

Existence is propositional truncation in Unit. No definition through prime ideals is used, and $N = 0$ is allowed.

Lemma 3. *For C as above and every $a \in C$, there is merely a $b \in C$ with $a = a^2b$. Consequently C is reduced and zero-dimensional.*

Proof. Every k-algebra is k-flat by the equational criterion. This does not require a basis: for a finite relation $\sum \alpha_i t_i = 0$, either all α_i vanish or one is invertible. In the second case the columns $e_i - (\alpha_i/\alpha_p)e_p$ explain the relation.

Apply Proposition 2 to $C \to C/(a)$. This quotient is C-flat. The checked flat-kernel lemma gives $a = az$ with $z \in (a)$; writing $z = ab$ yields $a = a^2b$. This is von Neumann regularity, which implies both assertions. For example, if $a^m = 0$, iterating $a = a^2b$ gives $a = a^m b^{m-1} = 0$. □

This removes the rational-function derivation extension from Lombardi's Theorem 4.5. It also removes primitive elements, perfection and discriminants from the *finiteness* stage. It uses neither ZMT nor the theorem flat plus unramified implies étale. Its dependencies are the explicit diagonal, base-change, and retraction arguments above.

The finite-span step can be simplified further than the initial version of this note. Induction on *algebra generators*, rather than on a chosen finite presentation, avoids contraction ideals entirely.

Proposition 4. *A finitely generated zero-dimensional algebra over a discrete field is finite. If it is finitely presented, it also has a finite basis.*

Construction. Induct on the number n of algebra generators $x_1, \ldots, x_n$ of C. For $n = 0$, the single element 1 spans C over k, including when C is the zero algebra. No decision between $C = 0$ and $C = k$ is needed.

For $n > 0$, zero-dimensionality supplies $x_1^N(1 - x_1 b) = 0$. Write $b = P(x_1, \ldots, x_n)$, using algebra generation. The polynomial

$$f = X_1^N(1 - X_1 P)$$

vanishes in C and is nonzero: its coefficient at $X_1^N X_2^0 \cdots X_n^0$ is 1. This also works when $N = 0$. A nonzero constant relation gives the zero algebra, which is finite. Otherwise a triangular change of variables of Nagata type makes f monic in the last variable T, after dividing by its nonzero leading scalar. With every exponent less than M, the substitutions $X_i = Y_i + T^{M^i}$ for $i < n$, keeping $X_n = T$ fixed, give distinct top T-degrees to the original monomials. This works over small fields.

Let $A' \subseteq C$ be the image subalgebra generated by the images of $Y_1, \ldots, Y_{n-1}$, and let t be the image of T. The inverse coordinate substitution shows $C = A'[t]$. The monic equation of degree d makes $1, t, \ldots, t^{d-1}$ span C over A', so C is integral over A' by the determinant trick.

Zero-dimensionality descends along this integral injection. Explicitly, for $x \in A'$, suppose

$$x^N = bx^{N+1}, \qquad b^h + c_{h-1}b^{h-1} + \cdots + c_0 = 0,$$

where the c_i belong to A'. The elementary *power absorption* lemma, proved by induction, is

$$x^N = b^j x^{N+j} \qquad (j \geq 0).$$

Consequently $b^j x^{N+h} = x^{N+h-j}$ for $0 \le j \le h$. Multiplying the monic equation by x^{N+h} therefore gives

$$x^N(1 + c_{h-1}x + \cdots + c_0 x^h) = 0.$$

Injectivity makes this an equation in A'. That subalgebra is generated by $n-1$ elements, so induction makes it finite over k. Combining the two finite spanning families proves that C is finite over k. Neither a finite presentation of A' nor generators of a contraction ideal are needed.

For the additional basis conclusion, use the finite presentation

$$C = k[X_1, \ldots, X_n]/(f_1, \ldots, f_m).$$

To obtain a basis, do not merely assume that a finitely generated k-module has one. Use the finite spanning family and the determinant trick to obtain monic annihilating polynomials $p_i(X_i)$ for the original generators. Then

$$Q_0 = k[X_1, \ldots, X_n]/(p_1(X_1), \ldots, p_n(X_n))$$

has its explicit bounded-monomial basis. The original equations f_i generate an ideal of Q_0 with finite k-spanning family consisting of their products with those basis monomials. Gaussian elimination on this finite matrix constructs a basis of the quotient C. This last step uses the original finite presentation essentially. □

The earlier contraction route remains valid: coherence of a finite free polynomial module gives generators of a contraction by projecting the last coordinates of the syzygies of $[V, -e_0]$. It is unnecessary in the simplified finite-span proof above. The primitive-element stage is now checked as described below. The Nagata substitution, generated image subalgebra, integral descent and finite-span induction now all check in `sag/etale-field-finite` and `sag/etale-field-generated`. Their combined theorem is `ZeroDimensionalFinite`.

The application `UnramifiedFieldFiniteGeneral` takes just finite presentation and formal unramifiedness over a discrete nonzero field.

The separate basis assembly also checks as `FinitePresentedFieldBasis`. It constructs the iterated monic cover, proves that the original equations generate its kernel, and forms the finite matrix consumed by quotient Gaussian elimination. Neither that matrix nor the cover's universal property is assumed. The file named `WeakNullstellensatz` in the affine tree is an SQC statement; it is not this finiteness theorem.

5 A small trace identity after finiteness

This implication now checks as `FiniteUnramifiedTrace` in `sag/etale-field-trace`, with an explicit finite basis as input. The scalar-extension algebra and its coordinates in the algebra tensor product are constructed; no tensor-basis hypothesis is assumed.

With a basis $b_1, \ldots, b_d$ of C, write the diagonal element as $\epsilon = \sum_i b_i \otimes c_i$. Let λ_i be the coordinate functions. Comparing coefficients in the equalisation identity gives, for every $z \in C$,

$$zc_i = \sum_j \lambda_i(zb_j)c_j.$$

Put $z = ab_i$ in the ith identity and sum over i. The decisive coordinate calculation is

$$\sum_i \lambda_i(ab_ib_j) = \sum_i \lambda_i((ab_j)b_i) = \mathrm{Tr}_{C/k}(ab_j),$$

since these are the diagonal entries of multiplication by ab_j. Since $\sum_i b_i c_i = 1$, exchanging the two finite sums gives

$$a = \sum_j \mathrm{Tr}_{C/k}(ab_j)c_j.$$

Hence the trace-pairing matrix has an inverse; its determinant is a unit. This is the $\mu(\epsilon) = 1$ specialisation of the trace identity in Quitté–Lombardi [3, Theorem 3.1]. Only this specialisation is needed, not the full norm–different divisibility theorem. The diagonal certificate has also avoided the detour through a perfect closure in [1, Theorem 4.3].

6 Selected linear algebra and simple roots

The relative-basis step can be confined to one specified decidable prime of a regular ring A. Start with a finite presentation $A^m \to A^n \to M \to 0$. The quotient $A/\mathfrak{p}$ is a discrete field: a regular witness $a = a^2 b$ gives the inverse of $\bar{a}$ when $a \notin \mathfrak{p}$. Run ordinary Gaussian elimination there, and lift only its finite projection, section and relation-coefficient matrices. Their equation errors form a finite family in $\mathfrak{p}$.

For an error $a \in \mathfrak{p}$, a regular witness supplies the idempotent $1 - ab \notin \mathfrak{p}$ which kills a. The product of these finitely many idempotents kills every matrix error. Thus on one selected factor eA, the lifted matrices give exact quotient coordinates and an actual basis of eM. This construction is checked in `sag/etale-field-selected-gaussian`; its algebra-basis adapter is `sag/etale-field-selected-basis`. No full decomposition of A is constructed.

There is also a smaller replacement for the discriminant calculation which makes a monogenic polynomial separable. For a finite algebra A with nondegenerate trace and a k-derivation $D : A \to A$, the commutator identity

$$[bD, m_a] = m_{bD(a)}$$

gives $\mathrm{Tr}(bD(a)) = 0$ for every b, hence $D(a) = 0$. For $Q = k[T]/(f)$, an annihilator h of $f'(T)$ defines the root $T + h\varepsilon$ in the dual numbers and therefore an actual derivation with $D(T) = h$.

Only trace nondegeneracy on a selected factor eQ is needed. Construct the relative derivation $Q \to eQ$ and restrict it to eQ. The equation $D(e) = D(e) + D(e)$ gives $D(e) = 0$, without dividing by 2, and $D(eT) = D(T) = h$. Nondegeneracy makes $h = 0$. Finally regularity $f' = (f')^2 b$ and the vanishing of the annihilator give $f'b = 1$ on this factor.

These steps now check. For an element $x \in C$, the constructed minimal quotient $Q = k[T]/(f)$ embeds in C. Selected relative bases, rectangular trace cyclicity, and trace transitivity pass nondegeneracy first to the selected factor of C and then to eQ. The result in `sag/etale-field-selected-separable` is a monic polynomial which annihilates x globally and has invertible derivative on a factor outside the specified prime. For a finite family of x's, multiply their independently constructed idempotents; every derivative inverse equation remains valid. This does not assert that the original polynomials are separable on their discarded factors.

7 The selected primitive-element proof

LQ VI–1.4, VI–1.7 and VI–1.9 give separable monogenic factors from the finite basis and invertible trace pairing. The preceding selected trace argument supplies the simple-root part. A further simplification of the large-field choice uses the diagonal directly, as described below; no irreducible factorisation over an arbitrary discrete field is being assumed.

For the local theorem, carry the specified prime through each idempotent refinement. Decidability and primality select the side containing the prime; products of these idempotents retain it. The checked construction does not retain the discarded factors or assemble a global product decomposition.

The small-field branch remains necessary. For example $\mathbf{F}_2^3$ is unramified and finite but is not monogenic: every element satisfies $X^2 - X$, so the subalgebra it generates has dimension at most two. At a specified prime, however, one coordinate factor is $\mathbf{F}_2$ and is monogenic.

LQ's large-field argument chooses a good linear combination by a nonzero discriminant polynomial. Its alternative branch works over the finite field generated by the relevant coefficients. A useful implementation of this dichotomy is bounded closure: grow a finite set containing $0, 1$ and those coefficients, closing under addition, negation, multiplication and total inverse (with $0^{-1} = 0$), until it is a finite subfield or has more elements than the required bound. Including inverses avoids a separate finite-injection argument for the subfield. This construction now checks as `FiniteCoefficientField`; it does not decide whether the whole field is finite.

The finite-field primitive-element argument is now checked as well. Choose an element of maximal multiplicative order by finite search. An explicit gcd-stripping construction combines two orders into an order divisible by both. Maximality then makes the selected order a period of every unit. Its distinct powers, together with any alleged missing nonzero element, would give too many roots of $T^m - 1$. The polynomial root bound excludes that case. The common monic annihilator $T^q - T$ has derivative -1; both assertions are proved from the finite enumeration. This uses no field classification or irreducible factorisation.

The small-field assembly constructs the finite subalgebra generated over the actual finite coefficient field, selects its field factor, and applies this primitive-element theorem. Its inclusion in the ambient algebra sends 1 to the selected idempotent e. The checked `CornerHorner` identities retain that factor in both values and derivatives. The resulting `SmallFieldBranch` produces exactly the selected certificate required by the local proof, from the finite coefficient field and the finitely many monic equations. These equations are themselves constructed by the determinant trick from the finite basis. Their lower coefficients are collected in one finite family before applying bounded closure. In the small branch, each occurrence has an explicit lift to the resulting finite subfield, so the actual monic lists and their root equations are available there.

A diagonal route for the large-field choice

Here is the checked replacement for the universal splitting-algebra and discriminant calculation. Write $D = C \otimes_k C$ and $\Delta x = x \otimes 1 - 1 \otimes x$. For a basis $b_1, \ldots, b_n$, the diagonal certificate implies that ϵ and the Δb_i generate the unit ideal of D. Consider

$$g(T) = \epsilon + \sum_{i=1}^{n} \Delta b_i T^i \in D[T].$$

Its coefficients generate the unit ideal. Polynomial McCoy therefore makes multiplication by g injective. On the constructed finite $k[T]$-basis of $D[T]$, matrix McCoy makes its determinant a nonzero polynomial. Its degree is at most $n \dim_k D = n^3$. Sufficiently many distinct scalars and the root bound supply an a where the determinant is nonzero, making $g(a)$ invertible. For $z = \sum_i a^i b_i$, this gives

$$(\Delta z)w = 1 - \epsilon.$$

The determinant, polynomial-basis and specialisation assembly for this search is checked in `sag/etale-field-diagonal-separator`. It requires $n^3 + 1$ distinct scalars and constructs both z and w.

The implication from this certificate to the needed coordinate equations is checked as well. If $p : C \to T$ is an algebra map and a k-linear map $\lambda : C \to T$ satisfies $\lambda(zc) = p(z)\lambda(c)$, contract the left tensor factor by λ. This kills every multiple of Δz. Since

$$\Delta c = (\Delta z)(w\Delta c),$$

it gives $\lambda(c) = \lambda(1)p(c)$. Apply this to the coordinates of a selected relative basis over $k[z]$. Expanding 1 selects one coordinate $\lambda_j(1)$ outside the prime. Regularity makes it invertible on a further selected factor, where the displayed equation expresses every element as an element of $k[z]$. The coordinate and monic-representative constructions now check; they avoid a separate relative-rank-one argument.

The final assembly in `sag/etale-field-selected-complete` applies bounded coefficient closure with this scalar bound. Its two branches return the same selected certificate. In the large branch the generation and simple-root idempotents are intersected; in the small branch the finite subfield and its lifted monic equations feed the finite-field construction. All finite choices remain under truncation. Neither a large-field hypothesis nor a decision that the whole field is finite is required.

8 The full product theorem and a shorter large-field argument

The checked endpoint is now

```
UnramifiedFieldProduct.Over[k,setk,field]
    .At[C,fp,unique].theorem.
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It merely constructs

$$C \simeq \prod_i k[T]/(f_i), \qquad f_i \text{ monic}, \quad f_i' \text{ invertible modulo } f_i.$$

Both directions are algebra maps with proved inverse equations. The finite product is represented by an ordinary finite binary tree. Zero factors are allowed, represented by the monic polynomial 1. No irreducibility is asserted: these are separable monogenic algebras, and decomposition into fields would require additional factorisation information about k.

8.1 A tensor separator gives a global generator

Suppose the scalar search has produced $x \in C$ and $w \in C \otimes_k C$ with

$$(x \otimes 1 - 1 \otimes x)w = 1 - \varepsilon.$$

Let f be the minimal monic annihilator of x, of degree $m > 0$. Put

$$H(X, Y) = \frac{f(X) - f(Y)}{X - Y}.$$

This notation denotes a polynomial constructed by Horner recursion; it is not division in the tensor algebra. Since $f(x) = 0$, the separator equation and diagonal equalisation give

$$H(x \otimes 1, 1 \otimes x) = \varepsilon H(x \otimes 1, 1 \otimes x) = \varepsilon(1 \otimes f'(x)).$$

Minimality makes x^{m-1} independent of the preceding powers. The existing Gaussian quotient constructs a k-linear functional $\lambda : C \to k$ which is zero on $1, x, \ldots, x^{m-2}$ and takes x^{m-1} to 1. The polynomial H is monic of degree $m-1$ in its first variable. Contracting that tensor factor therefore gives

$$1 = f'(x)(\lambda \otimes 1)(\varepsilon).$$

This explicitly constructs the derivative inverse. No regularity is needed for this step.

Let $Q = k[T]/(f)$. Minimality embeds Q in C. It is finite and reduced, hence regular by the checked theorem. An injective map from a regular ring reflects units: if $a = a^2b$, the complement $1-ab$ annihilates a, and its image must vanish when the image of a is a unit. Thus $f'(x)^{-1}$ belongs to Q.

The coordinates of $H(x \otimes 1, 1 \otimes x)$ in the left tensor basis belong to Q. The formal proof constructs their preimages by Horner recursion using the multiplication matrix of x. The same is then true of the coordinates c_j of ε, since $\varepsilon = (1 \otimes f'(x)^{-1})H$. Finally

$$a = \sum_j \mathrm{Tr}(ab_j)c_j$$

constructs a preimage in Q of every $a \in C$. Consequently $Q \to C$ is an isomorphism. Degree zero is handled by the equation $1 = 0$. This argument replaces selected relative bases and further idempotent refinements in the large-field branch.

8.2 All small-field factors, with finite termination

In the other branch, the actual finite coefficient field k_0 and the lifted monic equations construct a finite reduced subalgebra $A = k_0[b_1, \ldots, b_n] \subseteq C$. Enumerate all elements of A. For each element a choose a regular witness $a = a^2b$, inside the finite truncation, and split every current piece by ab and $1-ab$. On one piece a is invertible; on the other it is zero. Previously resolved equations survive further refinement. Induction on the enumeration length supplies termination, without dimension descent.

Every nonzero final corner eA is thus an explicitly enumerated finite field. Its checked primitive-element construction represents each eb_i by a polynomial over k_0. These representations span eC over k and make it monogenic. Trace nondegeneracy passes to eC and then to its minimal-polynomial quotient, giving separability by the checked monic trace theorem. Zero pieces are included explicitly. The isomorphisms $eC \simeq efC \times e(1-f)C$ assemble the resulting finite product. Both sides of every split are retained; no prime is chosen.

9 Checked milestones and remaining local integration

1. **The diagonal and regularity.** Checked: the presented diagonal, $J = J^2$ directly from `Unique`, the determinant construction, general flat base change and averaging, and $a = a^2b$ over a discrete field. The exported `UnramifiedFieldRegular` theorem also proves reducedness and algebraic zero-dimensionality. It assumes only the stated field, finite-presentation and formal-unramifiedness data, with no assumed transfer principle or ZMT.

2. **Finite basis.** The finite-span part of Proposition 4 is checked, using monic substitutions, induction on algebra generators and integral descent. This already supplies the finite-fibre input needed before ZMT. The separate basis construction is also checked, using bounded monic quotients and quotient Gaussian elimination. It does not require a general dimension theory or a separate Gröbner-basis development.

3. **One separable monogenic neighbourhood.** The trace identity is checked with a finite basis as input. The selected relative-basis, simple-root, small-field and large-field constructions are checked, together with their bounded-coefficient dichotomy. The final theorem `UnramifiedFieldSelected.Over.At.theorem` takes only a discrete nonzero field, a finitely presented formally unramified algebra, and a specified decidable proper prime. It merely supplies a finite basis and the exact `SelectedFactorStatement.Selected` certificate. It assumes neither ZMT nor a field structure theorem.

4. **The full product structure.** `UnramifiedFieldProduct` now gives the product above from the same field, finite-presentation and formal-unramifiedness hypotheses, with no prime input. A fresh check gives 18757 declarations, 0 failed, in 9.42 s; 847 declarations in 25 modules are new. All 446 modules pass `segc all`. Regressions include the base field $\mathbb{F}_2$, the constructed standard algebra for T^2-T, the actual large-field separator, $\mathbb{F}_2^3$ through the small branch, and the zero algebra. The product theorem is not yet wired into a replacement for the existing selected-factor local adapter.

5. **The residual bridge.** Base-change `Unique` to $R/\mathfrak{m}$, transport the chosen factor through the ZMT localisation, and lift its finitely many coefficients and generation equations under truncation. Use the weaker finite-generator interface described above. Check the nonreduced source $B/\mathfrak{m}B$ and the empty/zero cases explicitly. In particular, do not assume $B/\mathfrak{m}B$ injects into the localised special fibre: each equality pulled back from that fibre must carry an explicit power of the localisation element s. The base change, the smaller local interface and the generic clearing lemmas now check. `QuotientPowerClearing` proves

$$\eta(b) = 0 \text{ in } (B/I)[1/s] \quad \Longleftrightarrow \quad \exists N\ s^N b \in I.$$

   If $s^N(1 - sv) \in I$, it also constructs mere representatives in B for every element of the localisation: replace b/s^n by bv^n. This gives surjectivity, with no claim of injectivity.

   `LocalResidueField` constructs the discrete nonzero field $R/\mathfrak{m}$. `QuotientFibreBaseChange` equips the actual quotient $E/\mathfrak{m}E$ with the mapped finite presentation and transfers formal unramifiedness to it. `FiniteSpecialFibre` consequently supplies the usual quotient-module finiteness hypothesis of ZMT. The assembly of a supplied selected factor's finitely many coefficients and identities into the exact local interface now checks in `sag/etale-field-selected-lift`. It still takes the selected field certificate, compatible prime, quasi-inverse relation and local ZMT data explicitly. Its split-algebra regression includes a nonzero quotient class which vanishes after localisation. The subsequent adapter in `sag/etale-field-local-complete` now uses `UnramifiedFieldSelected` to construct the basis and selected certificate. It retains explicit finite presentation and formal unramifiedness of the exact localized fibre, the field structure of the actual residue quotient, and the compatible prime, quasi-inverse and local ZMT data.

The existing tensor product of presented algebras (`sag/tensor/Tensor.seg`), effective quotients, determinant trick, flat retractions, principal base change, FieldCFN and HilbertCFN are reused throughout the checked construction. The existing local theorem is unchanged. Its final unconditional application still requires the connections supplying the compatible prime, quasi-inverse relation and local ZMT data on the actual localized special fibre.

References

[1] H. Lombardi, *Un théorème décisif*, 2025, arXiv:2506.07098. Local source: `CANONICITY/decisif.pdf`; especially Theorems 4.3–4.5.

[2] H. Lombardi and C. Quitté, *Commutative Algebra: Constructive Methods*, 2015. Local source: `COLL/LQ.pdf`; III–9, IV–8.13–8.16, VI–1.4–1.9, VII–1.4–1.5. The numbering here is that of this English PDF.

[3] C. Quitté and H. Lombardi, *Identités algébriques permettant de démontrer qu'une algèbre libre finie nette est traciquement étale*, 2025, arXiv:2506.03851, Theorem 3.1.

[4] The Stacks Project, Tag 02FL, the diagonal idempotent for an unramified algebra.